Name (print):_____Section (day/time):_____

ME139L-Experimental Heat Transfer
Pre-Lab Assignment for Lab 2

Problem 1: In this lab, you will use infrared thermography to measure the temperature distribution in a composite rod consisting of two segments made of different materials, as illustrated in Figure 1. The thermal conductivity of one of the segments is known based on a different measurement method. You will be asked to use the obtained temperature distribution to determine the unknown thermal conductivity of the other rod material, the contact thermal resistance between the two segments, and the surface heat loss. Before conducting these experiments you should have some ideas of the temperature variation expected along the rods, and how the thermal conductivity, thermal contact thermal resistance, and surface heat transfer coefficient affect the obtained temperature distribution.

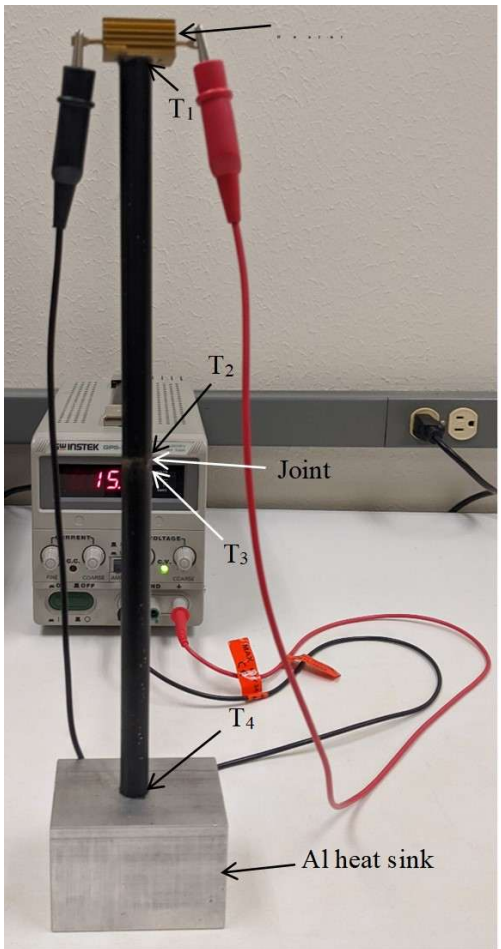


Figure 1: Composite rod setup for measuring thermal conductivity

Since the rod surface is not insulated, the surface heat loss cannot be ignored. The temperature distribution in the rod is non-linear because the heat flux changes throughout the entire length as heat is dissipated from the rod surface. In this case each segment of the rod can be modelled as a one dimensional fin where the base and tip of the fin are held at different and constant temperatures. According to this model, the temperature distribution in each rod segment is described by the following equation:

$$T(x) = \frac{(T_L - T_\infty)sinh(mx) + (T_b - T_\infty)sinh\big(m(L-x)\big)}{sinh(mL)} + T_\infty$$

where the tip temperature T_L can be taken as T_2 for the top segment and T_4 for the bottom segment, the base temperature T_b can be taken as T_1 for the top segment and T_3 for the bottom segment, T_∞ is the temperature of the ambient air, and $m \equiv (hP/k\,A_c)^{1/2}$is a fin parameter that depends on the thermal conductivity (k, surface heat transfer coefficient (h), cross sectional area (A_c), and perimeter (P) of the cross section of the rod.

Figure 2 shows temperature data taken by a teaching assistant (TA) for a composite rod. The open circles are the raw data, the solid lines are best fit curves to the data using the temperature distribution model described above, and the dotted lines indicate the slope of the temperature distribution at the junction.

Name (print):_____Section (day/time):_____

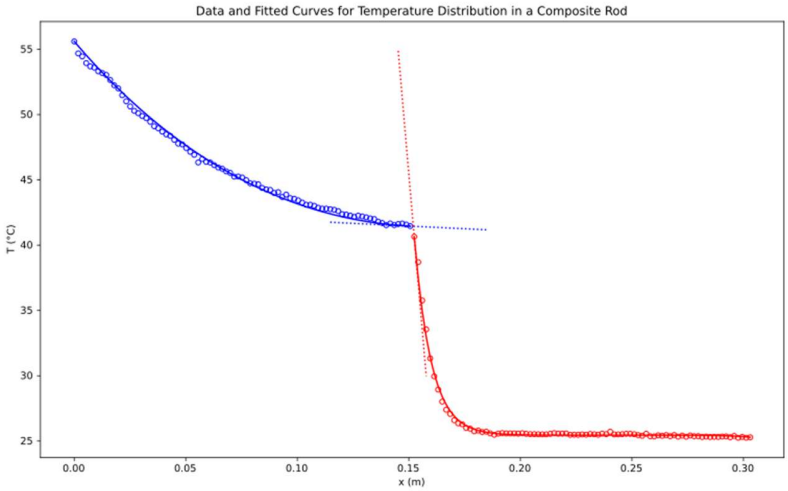


Figure 2: Data collected by a TA and fitted curves

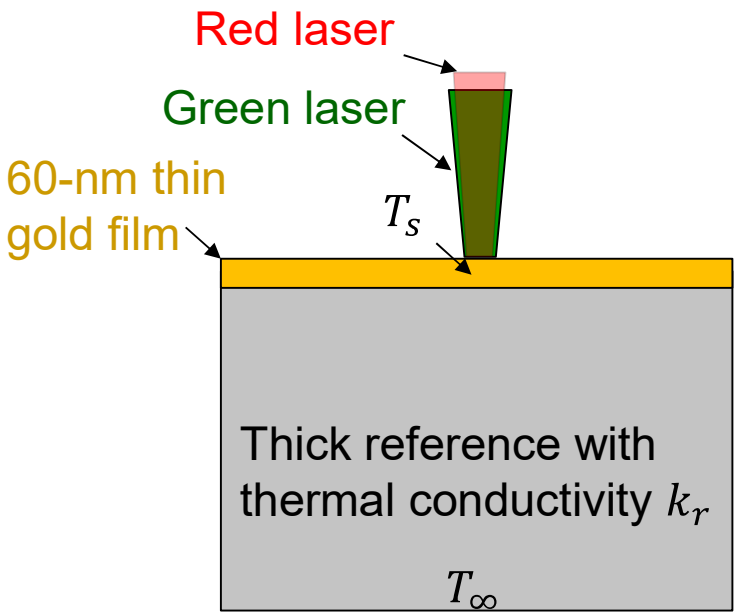
Based on the curve fitting algorithm, the optimum fitting parameters are m = 7.865 m^{-1} for the top segment and m = 130.4 m^{-1} for the bottom segment. In addition, the IR camera measures T_1 = 55.60°C, T_2 = 41.46°C, T_3 = 40.65°C, and T_4 = 25.45°C. T_∞ is 25.45°C based on room air temperature measurements with a thermocouple. The length of each segment is L = 6 in = 0.1524 m. The thermal conductivity of the top segment is 109 W/m·K.

(a) Compute the slope of each temperature distribution at the junction between the two rods. Use these temperature gradients and the known thermal conductivity of the top segment to calculate the thermal conductivity of the bottom segment.

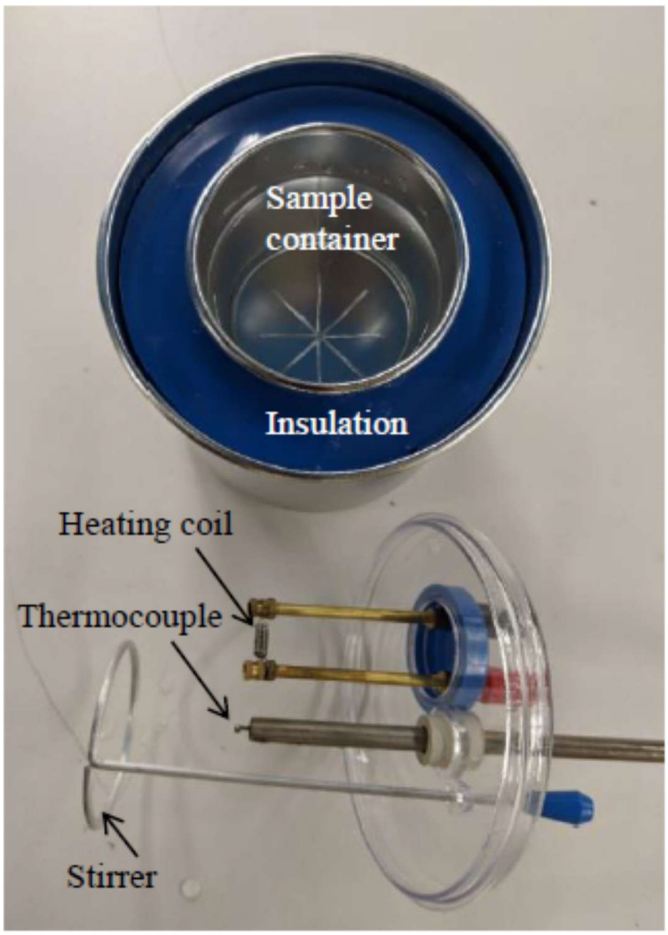
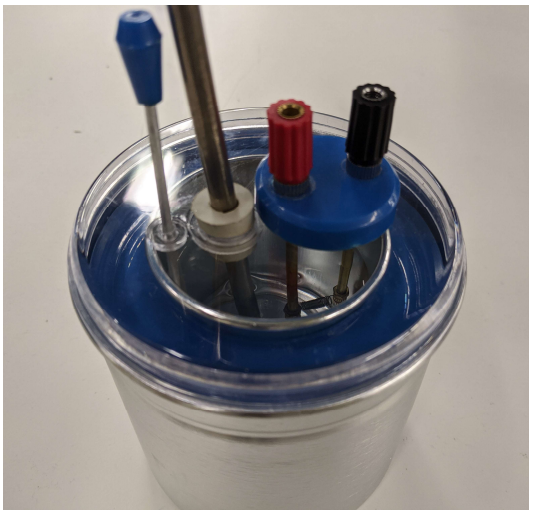
(b) Suppose we instead estimate the slope at the junction by just choosing a few data points near the junction and fitting a line to them. How will the estimate of each slope by this method compare to the values computed in Part a? Will we underestimate or overestimate the thermal conductivity of the bottom segment?

Name (print):_____Section (day/time):_____

Problem 2: The schematic below illustrates a new thermorefletance measurement experiment that has recently built by the teaching assistants and instructor for this lab. In this setup, a 532-nm-wavelegnth green laser beam is used to heat a 60-nm-thick gold film coating of a very thick and large plate sample. Another 640-nm-wavelegnth red laser beam is used to measure the reflectivity change of the gold film due to heating by the green laser. The gold film absorbs about 60% and reflects the rest of the incident green laser. The diameter of the focused green laser beam spot is about 10 μm at the gold film surface. The thermal conductivity of a thick glass sample is close to 1.3 W/m·K. Lateral heat conduction along the very thin gold film away from the green laser spot is negligible. The thermal resistance across the gold film thickness and its interface with the glass sample is negligible compared to the thermal resistance of the thick glass sample. Based on the conduction shape factor for the case of a heated circular disk on a semi-infinite medium (Case 10 of Table 4.1 of the textbook, Fundamentals of Heat and Mass Transfer, 8th Edition, Bergman, Lavine; see lecture 3 slides), calculate the incident green laser power that is needed to cause 1°C increase of the gold film temperature in the green laser spot.



Name (print):_____Section (day/time):_____

Problem 3: An electrical calorimeter contains a cylindrical sample container with an inner diameter of 6 cm and height of 7.4 cm, a 4Ω coil heater, a K-type thermocouple, and a stirrer. The cylindrical container is thermally insulated on the bottom and the cylindrical surface of the container. The container is partly filled with 150 milliliters of water. After the heating coil and the thermocouple assembly is inserted into the water, a 9V voltage is applied to the heating coil.

a) In order to make the lumped capacitance assumption, what is the maximum allowed surface heat transfer coefficient on the exposed top surface of the water when heat loss through the thermal insulation at the bottom and cylindrical surfaces is negligible?
b) Determine the temperature rise of the water over a period of 1.5 minutes for the case when the top surface heat transfer coefficient is at this maximum allowed value.
c) Determine the temperature rise of the water over a period of 1.5 minute for the case when the top surface heat transfer coefficient is zero.